Yuqun Wu

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Education

University of Illinois at Urbana-Champaign Champaign, USA
Doctor of Philosophy in Computer Science *Aug 2023 - Present*
- Advisor: *Prof. Derek Hoiem*, *Prof. Shenlong Wang*

Master of Science in Computer Science (thesis) *Aug 2022 - May 2023*
- Advisor: *Prof. Derek Hoiem*, *Prof. Shenlong Wang*

Bachelor of Science in Computer Science & Statistics *Jan 2020 - Dec 2021*
- Highest Honors at graduation, Dean's list for all years, GPA: 4.0/4.0

Sun Yat-sen University Guangzhou, China
Bachelor of Science in Mathematics *Sep 2016 - Dec 2019*

Research Experience

University of Illinois at Urbana-Champaign Champaign, USA

Render to Reason: Novel-View Semantic Prediction for Spatial Reasoning in VLMs *Aug 2025 - May 2026*
Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang - ***In Submission***
- Project targeted improving VLM spatial reasoning with 3D geometry features, based on the finding that standard spatial QA is largely addressable from visual features and language priors, leaving geometry underused.
- Proposed novel-view semantic rendering, an auxiliary task that requires both geometry and visual features, improving performance on VSI-Bench and ReVSI.

SceneDiff: A Benchmark and Method for Multiview Object Change Detection *Jan 2025 - May 2025*
Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang - ***ECCV 2026***
- Project targeted the problem of identifying regions that have changed between a pair of captures (images or videos) of the same scene at different times.
- Proposed a dataset and a new method for object change detection between a pair of captures (images or videos) of the same scene at different times.

TextRegion: Text-Aligned Region Tokens from Frozen Image-Text Models *Dec 2024 - May 2025*
Advisor: Prof. Derek Hoiem - ***TMLR 2025***
- Project proposed a training-free approach to enhance spatial understanding capacity of existing image-text models.
- Contributed to idea development, running experiments, and paper writing.

Region-based Representations Revisited *Sep 2023 - Nov 2023*
Advisor: Prof. Derek Hoiem - ***CVPR 2024***
- Project targeted at investigating new representation by combining SAM regions and dense features to solve various vision tasks, including semantic segmentation, object retrieval, video classification, and scene segmentation
- Responsible for implementation of feature extraction and pooling pipelines, and scene segmentation application on ScanNet

Improving Neural Radiance Fields with Patch-based Monocular Guidance *Jan 2023 - May 2023*
Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang - ***3DV 2025***
- Project aimed to create 3D models that provide accurate geometry and view synthesis, partially closing the large geometric performance gap between NeRF and traditional MVS methods
- Proposed appearance regularization of normalized cross-correlation (NCC) and structural similarity (SSIM) between randomly sampled novel and training view to improve general performance

Plenoptic PNG: Real-Time Neural Radiance Fields in 150 KB *Aug 2022 - Dec 2023*
Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang - ***3DV 2025***
- Project presented Quantized Fourier Features, which encoded a 3D scene into an extremely compact representation from 2D images and enabled its transmittance, decoding, and rendering in real-time across various platforms.

- Contributed to blending Quantized Fourier Features into different network setups, running experiments, and paper writing.

Sparse SPN: Depth Completion from Sparse Keypoints *Sep 2021 - Nov 2022*
Advisor: Prof. Derek Hoiem

- Project targeted fast point clouds reconstruction from single view depth completion with SfM inputs.
- Proposed a novel method that outperforms existing depth completion pipelines given sparse keypoint depth, and reconstructed complete point clouds given SfM setup

GRIT: General Robust Image Task Benchmark *Jun 2021 - Aug 2021*
Advisor: Prof. Derek Hoiem

- Rendered surface normal of object-centric and scene-centric datasets, and split them into training, validation, and testing sets
- Trained a baseline network with training sets, and compare it with several other pretrained state-of-the-art normal estimation networks with testing sets
- Challenge Organizer of the 2nd workshop on Open World Vision of CVPR 2022

University of California San Diego Remote

Lighting completion from sparse lighting samples *Jun 2022 - Sep 2022*
Advisor: Prof. Manmohan Chandraker

- Project aims at recovering per-pixel spatially-varying lighting maps taking single color image and sparse lighting samples
- Investigated 2D lighting completion methods with differentiable rendering and compare to pure RGB-based estimati -on networks

Industry Experience

Research Scientist Internship Redmond, USA
Meta *May 2025 - Aug 2025*
Research Scientist Internship Remote
Stability AI *May 2026 - Aug 2026*

Service

Conference Reviewer Champaign, USA
CVPR, ECCV, WACV *2024 - 2026*
Teaching Assistant Champaign, USA
University of Illinois at Urbana-Champaign *Aug 2022 - May 2023*

- Course: *CS 445 Computational Photography*, *CS 441 Applied Machine Learning*

Skills

- **Programming Languages**: Python, C/C++, JavaScript, R
- **Other Tools**: Git, Pytorch, Latex